

SMIT KETANKUMAR SHAH

M.S in Electrical System Engineering

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SUMMARY

Self-motivated Electronics & Telecommunication Engineer with a Master's degree in Electrical Systems Engineering and strong expertise in digital logic, analog circuits, VLSI, embedded systems, robotics, IoT, and power electronics. Skilled in hardware implementations, PCB design and circuit simulation, SQL, Tableau, and programming languages like Python and MATLAB.

SKILLS

Programming Languages:	C, C++, Arduino, Python, MATLAB, SQL
PCB Design:	KiCad, LTspice, Eagle, EasyEDA, Proteus, Altium, Cadence
VLSI:	Xilinx ISE, Xilinx Vivado, FPGA
Robotics:	RoboAnalyser, ROS, Gazebo
Embedded Systems:	Keil, MPLAB, RTOS
Power Electronics:	NI Multisim, NI Labview, Psim, Plecs, Simetrix

WORK EXPERIENCE

Sep. 2026 – Present
Master Thesis Student
Infineon Technologies AG, Warstein, North Rhine-Westphalia, Germany

- Currently conducting an industry-based Master's thesis on passive filter optimization for a 10 kW three-phase, 3-level NPC2 grid-tied inverter, building simulation models and running automated parametric studies to support filter design trade-offs and switching-frequency selection.

Mar. 2023 – Aug. 2026
Working Student: Technical Marketing and Application Engineering
Infineon Technologies AG, Warstein, North Rhine-Westphalia, Germany

- Designed and maintained PCB layouts for evaluation boards in Altium Designer and created simulation models for double-pulse testing.
- Developed Tableau dashboards and KPI reports using Denodo and Jama data to track project milestones, lead times, and requirements.
- Built CI/CD pipelines and automated workflows to extract product data, generate documentation, and publish reports to internal platforms.
- Conducted functional and validation testing on evaluation boards and IoT platforms, including automated double pulse testing with PyVerify.
- Supported competitive benchmarking by documenting products at PCIM Europe and developing tools for web data extraction and delta analysis.

Jul. 2021 – Sep. 2022
Assistant System Engineer
Tata Consultancy Services Ltd., Bangalore, Karnataka, India

- Contributed to the Cloud Migration project for Bed Bath and Beyond, migrating data from Teradata to GCP using BigQuery, SQL, and Python.
- Responsible for writing SQL queries, data validation, and Python coding for DAG creation. Optimized Teradata SQL queries to reduce execution time and implemented parallel processing to lower execution costs.
- Received recognition from Bed Bath and Beyond and was awarded TCS Star of the Month for exceptional performance.

Jan. 2020 – Jun. 2020
Student Intern
Makers Lab, Tech Mahindra, Pune, Maharashtra, India

- Worked on the company's one of the Big Bet projects named ROAD2 (Retrofit Option for Autonomous Driving – Level 2), focusing on the development of autonomous driving solutions.
- Responsible for circuit design, coding using Python, and testing to ensure the functionality and performance of the system.

PROJECTS

Sep. 2026 – Feb. 2027 (tentative)
Impact of Switching Frequency on Integrated LCL-EMI Filter Design: Sizing, Weight, and Cost Implications for NPC2 Grid-Connected Inverters
Currently conducting Master's research on the optimization of integrated LCL and EMI passive filters for a 10 kW, three-phase, 3-level NPC2 grid-tied inverter to address the lack of quantitative trade-off data in current literature. Developing a coupled electromagnetic model in PLECS/Simscape that accounts for both differential-mode and common-mode noise paths, while performing parametric sweeps across switching frequencies (20–100 kHz) to determine the minimum component values required for power quality and EMI compliance. Translating these electrical specifications into physical parameters (weight, volume) and economic metrics (BOM cost) using empirical sizing algorithms and manufacturer pricing models to generate Pareto fronts and identify the optimal switching frequency that minimizes the total passive component footprint.

Oct. 2023 – Sep. 2024
Rescue Robots
In this project, I was part of the Mapping Subgroup, which focused on improving map quality by enhancing robot localization, applying various optimization techniques, increasing loop closure reliability, and filtering dynamic objects. The project built on the LOAM algorithm, integrating IMU data and loop closure to improve map accuracy. I created a templated structure supporting multiple graph optimizers (g2o, SE-Sync) and implemented dynamic object filtering using SegFormer B1 for semantic segmentation. This approach involved projecting 3D Lidar points onto a 2D image plane to detect and remove dynamic objects. Through various test cases, I assessed the impact of IMU calibration, loop closure, and dynamic object filtering, achieving significant improvements in map quality.

Jun. 2020 – Jun. 2021
S2SLT-E2K: Sequence to Sequence Language Translation from English to Konkani using Neural Machine Systems
This was our B.Tech project. In this, we translated English sentences to Konkani sentences. We used an encoder-decoder LSTM layer in a sequence-to-sequence model and the Adam optimizer with various regularization parameters for training the model. We took approximately 70k pairs of sentences as a dataset for our project, and we were able to get an accuracy of around 60%.

Oct. 2019
Alcohol Detector using Arduino
In this project, we used an MQ-3 sensor and interfaced it with Arduino Uno, LCD, and Buzzer. I was responsible for circuit and PCB design and mounting the components. I was also heading the team. In the project, the MQ-3 sensor detects the proportion of Alcohol, and then if the proportion of alcohol is more than 40%, the LCD will display the alert that alcohol consumption is more, and the buzzer will make a sound.

Oct. 2019
PWM-based DC motor control using 8051
In this project, we used the 89C52 IC, which was interfaced with the motor driver to control the speed of the motor. We even built the power supply for driving the circuit with the help of a transformer, power diodes, and IC 7805 and 7905 to convert 230V AC to 5V DC. We were using a PWM signal to control the speed of the motor. I was responsible for circuit design and simulation in Proteus software and coding in Keil software. I was also heading this project.

EDUCATION

Oct. 2022 – Present
Universität Paderborn, Paderborn (Germany)
Master of Science in Electrical System Engineering
Specialization: Electronics and Devices

Jun. 2017 – Jul. 2021
Symbiosis Institute of Technology, Pune (India)
Bachelor of Technology, Electronics & Telecommunication Engineering

CERTIFICATIONS

Jan. 2026 Allegro X PCB Editor Basic Techniques v24.1 (Cadence Training Services)
Jul. 2024 Everybody's Introduction to Tableau (Linkedin Learning)
Dec. 2023 Altium Designer- Advanced (Altium Certification & Training)
May. 2023 Learning FPGA Development (Linkedin Learning)
May. 2023 Learning Verilog for FPGA Development (Linkedin Learning)
Mar. 2023 Learning Altium (Linkedin Learning)
Mar. 2021 AI for Everyone: Master the Basics (edX)
Sep. 2020 Hands-on workshop on Fundamental of Linux and Data Analysis by _VOIS
Jun. 2020 Programming for Everybody: Getting Started with Python (Coursera)
Jun. 2020 Self-Driving Cars Specialization (Coursera)
Mar. 2020 IC Design Process: A Beginner's Overview to VLSI Technology (Udemy)
Mar. 2020 KiCad (Udemy)
Feb. 2019 Programming with Python (Internshala)
Aug. 2018 Hands-on workshop on Arduino Programming by Symbiosis Institute of Technology, Pune

LANGUAGES

English - C1, **German** - B1, **Gujarati** - Native or Bilingual, **Hindi** - Native or Bilingual